$p_f - p_i = \Delta p =$ change in momentum of the object during the time interval $t_f - t_i = \Delta t$.

The integral $\int_{t_i}^{t_f} F\,dt = J$ is called the impulse and it is equal to change in momentum of the object.

If the force is constant over the time interval, then

$$\int_{t_i}^{t_f} F\,dt = F\int_{t_i}^{t_f} dt = F\left(t_f - t_i\right) = F\Delta t$$

$$F\,\Delta t = \Delta p \qquad (3.24)$$

Equation (3.24) is called the 'impulse-momentum equation'.

For a constant force, the impulse is denoted as $J = F\Delta t$ and it is also equal to change in momentum (Δp) of the object over the time interval Δt.

Impulse is a vector quantity and its unit is Ns.

The average force acted on the object over the short interval of time is defined by

$$F_{avg} = \frac{\Delta p}{\Delta t} \qquad (3.25)$$

From equation (3.25), the average force that act on the object is greater if Δt is smaller. Whenever the momentum of the body changes very quickly, the average force becomes larger.

The impulse can also be written in terms of the average force. Since Δp is change in momentum of the object and is equal to impulse (J), we have

$$J = F_{avg}\Delta t \qquad (3.26)$$

The graphical representation of constant force impulse and variable force impulse is given in Figure 3.21.

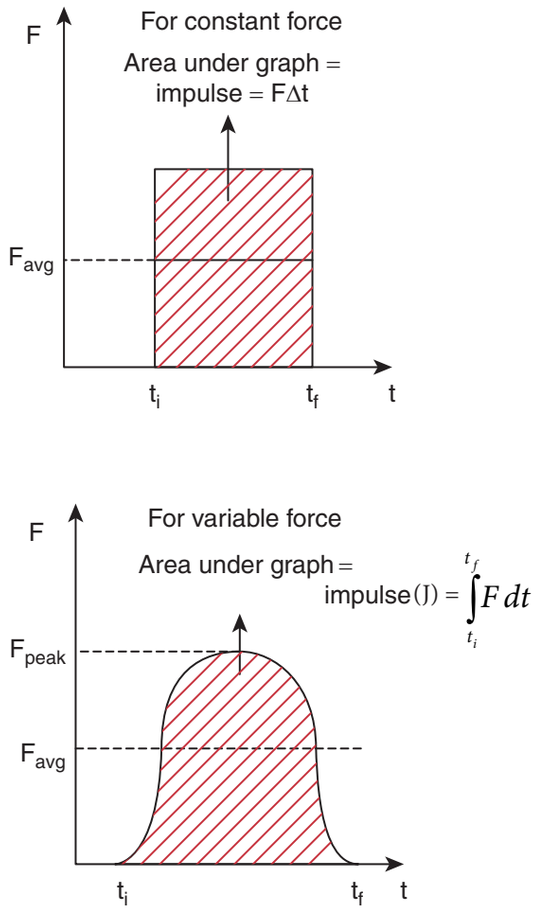


Figure 3.21 Constant force impulse and variable force impulse

Illustration

1. When a cricket player catches the ball, he pulls his hands gradually in the direction of the ball's motion. Why?

   If he stops his hands soon after catching the ball, the ball comes to rest very quickly. It means that the momentum of the ball is brought to rest very quickly. So the average force acting

UNIT-3(XI-Physics_Vol-1).indd 134 28-01-2020 13:48:23